Dire Dawa University Institute of Technology (DDUIoT)

College of Electrical and Computer Engineering (CECE)

Course: Data communication and computer Networking[ECEg-4201]

Chapter Five : Routing in Switched Networks

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Outline

- Concepts of circuit switching, packet switching
- Routing in Switched Networks
- Introduction to Routing.
- Graph Theory - Dijkstra Algorithm.
- Bellman Ford Algorithm
- Distance Vector, Link State

Circuit Switched Connection-oriented Networks

- To understand the challenge that the DoD researchers were faced with, it is necessary to look at how early telephone systems work.
- When a person makes a call using a traditional telephone set, the call first goes through a setup process, where all of the telephone switching locations between the person and the phone set that they are calling are identified.
- A temporary path, or circuit, is created through the various switching locations to use for the duration of the telephone call.
- If any link or device participating in the circuit fails, the call is dropped.
- To reconnect, a new call must be made, and a new circuit created between the source telephone set and the destination.

Continued…

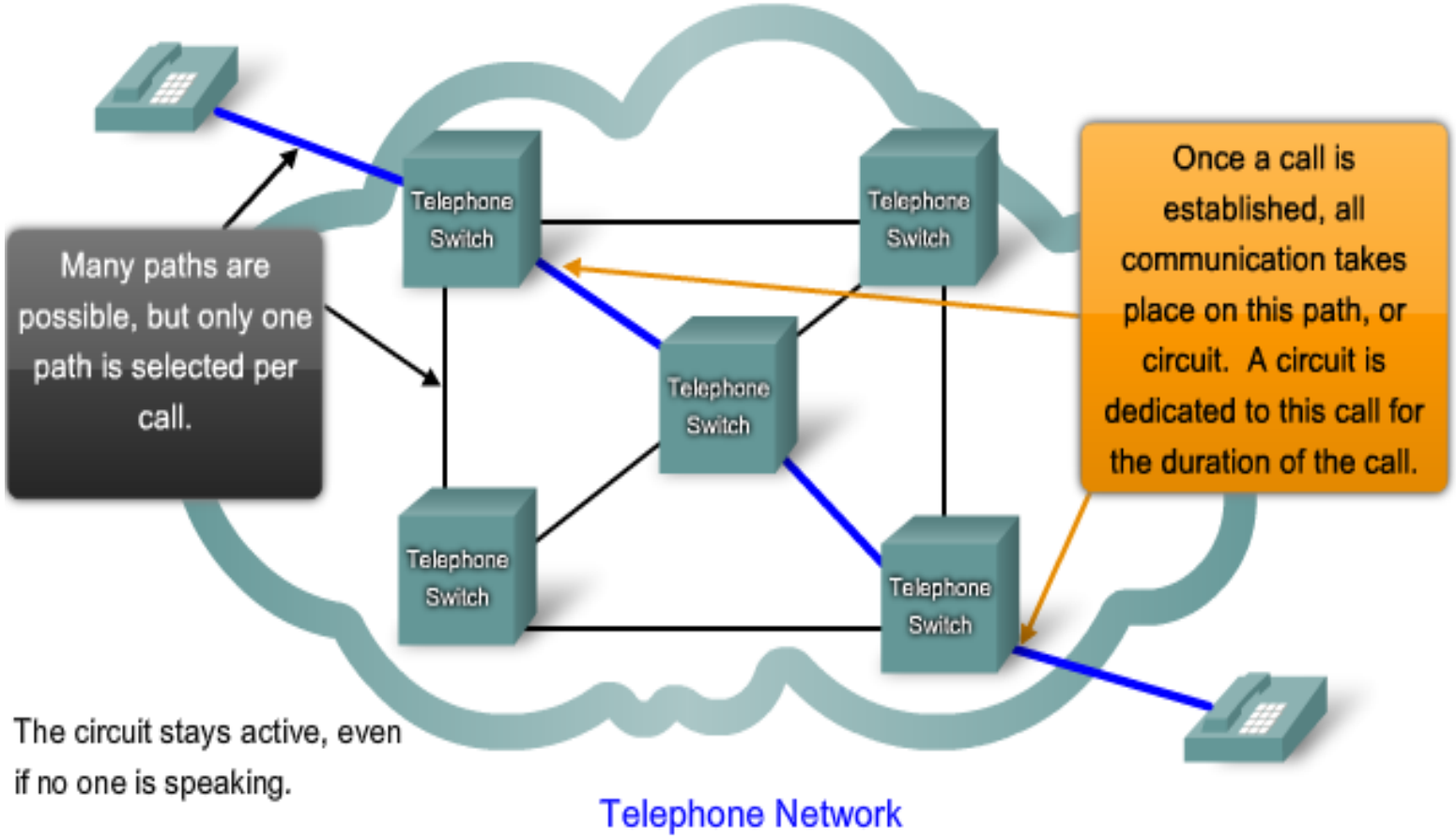
- Early circuit switched networks did not dynamically recreate dropped circuits.
- In order to recover from failure, new calls had to be initiated and new circuits built end-to-end.
- Many circuit switched networks give priority to maintaining existing circuit connections, at the expense of new circuit requests.
- In this type of connection-oriented network, once a circuit is established, even if no communication is occurring between the persons on either end of the call, the circuit remains connected and resources reserved until one of the parties disconnects the call.

Continued…

- Since there is a finite capacity to create new circuits, it is possible to occasionally get a message that all circuits are busy and a call cannot be placed.
- The cost to create many alternate paths with enough capacity to support a large number of simultaneous circuits, and the technologies necessary to dynamically recreate dropped circuits in the event of a failure, led the DoD to consider other types of networks.

❖ Circuit switched networks

Circuit Switching in a Telephone Network



Packet Switched Connectionless Networks

- In the search for a network that could withstand the loss of a significant amount of its transmission and switching facilities, the early Internet designers reevaluated early research regarding packet switched networks.
- The premise for this type of networks is that a single message can be broken into multiple message blocks.
- Individual blocks containing addressing information indicate both their origination point and their final destination.
- Using this embedded information, these message blocks, called packets, can be sent through the network along various paths, and can be reassembled into the original message upon reaching their destination.

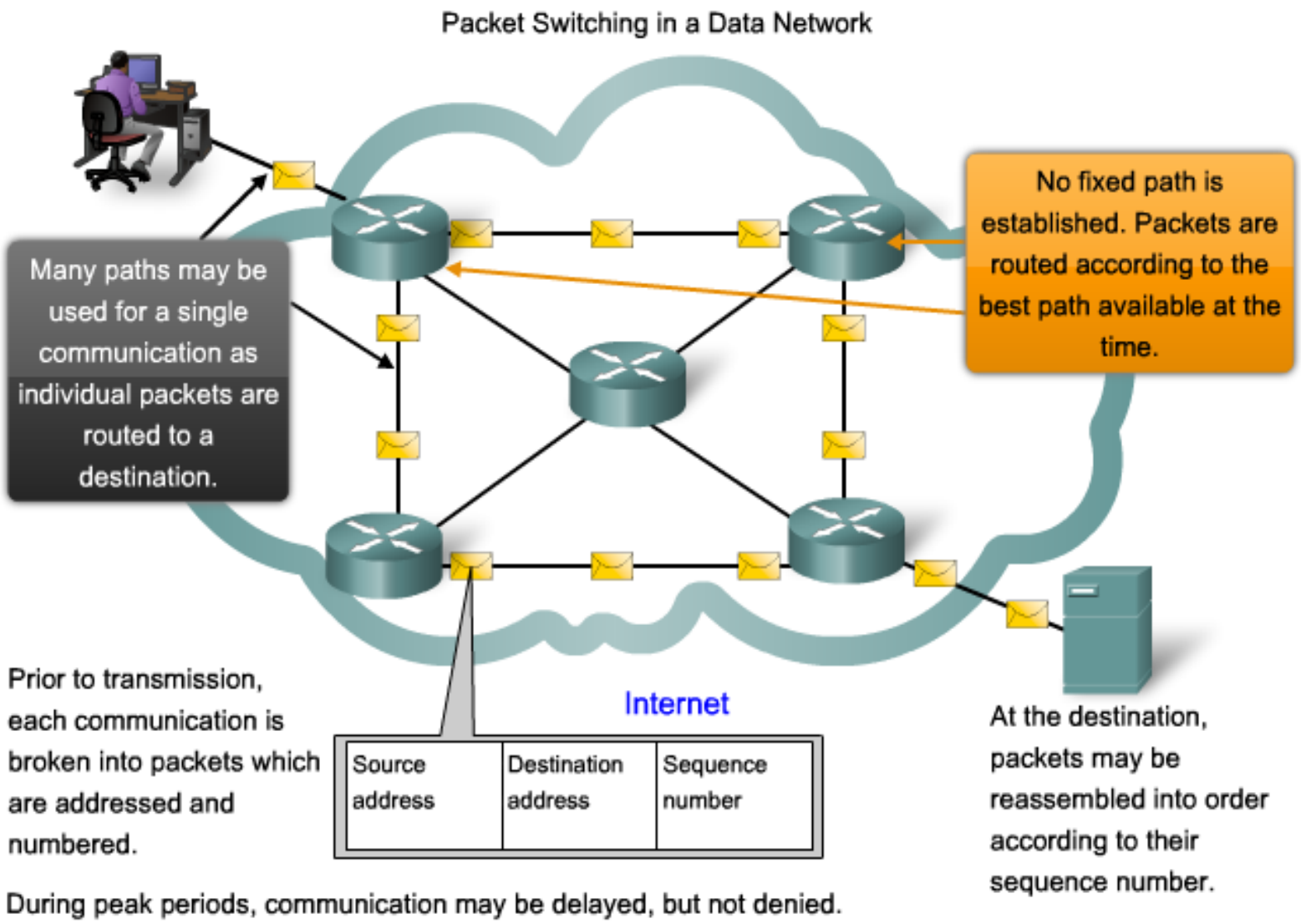
Utilizing packets

- The devices within the network itself are unaware of the content of the individual packets, only visible is the address of the final destination and the next device in the path to that destination.
- No reserved circuit is built between sender and receiver. Each packet is sent independently from one switching location to another.
- At each location, a routing decision is made as to which path to use to forward the packet towards its final destination.
- If a previously used path is no longer available, the routing function can dynamically choose the next best available path.
- Because the messages are sent in pieces, rather than as a single complete message, the few packets that may be lost in the advent of a failure can be retransmitted to the destination along a different path.
- In many cases, the destination device is unaware that any failure or rerouting has occurred.

Utilizing packets

- The DoD researchers realized that a packet switched connectionless network had the features necessary to support a resilient, fault tolerant network architecture.
- The need for a single, reserved circuit from end-to-end does not exist in a packet switched network.
- Any piece of a message can be sent through the network using any available path.
- Packets containing pieces of messages from different sources can travel the network at the same time.
- The problem of underutilized or idle circuits is eliminated -- all available resources can be used at any time to deliver packets to their final destination.
- By providing a method to dynamically use redundant paths, without intervention by the user, the Internet has become a fault tolerant, scalable method of communications.

❖ Packet switched network



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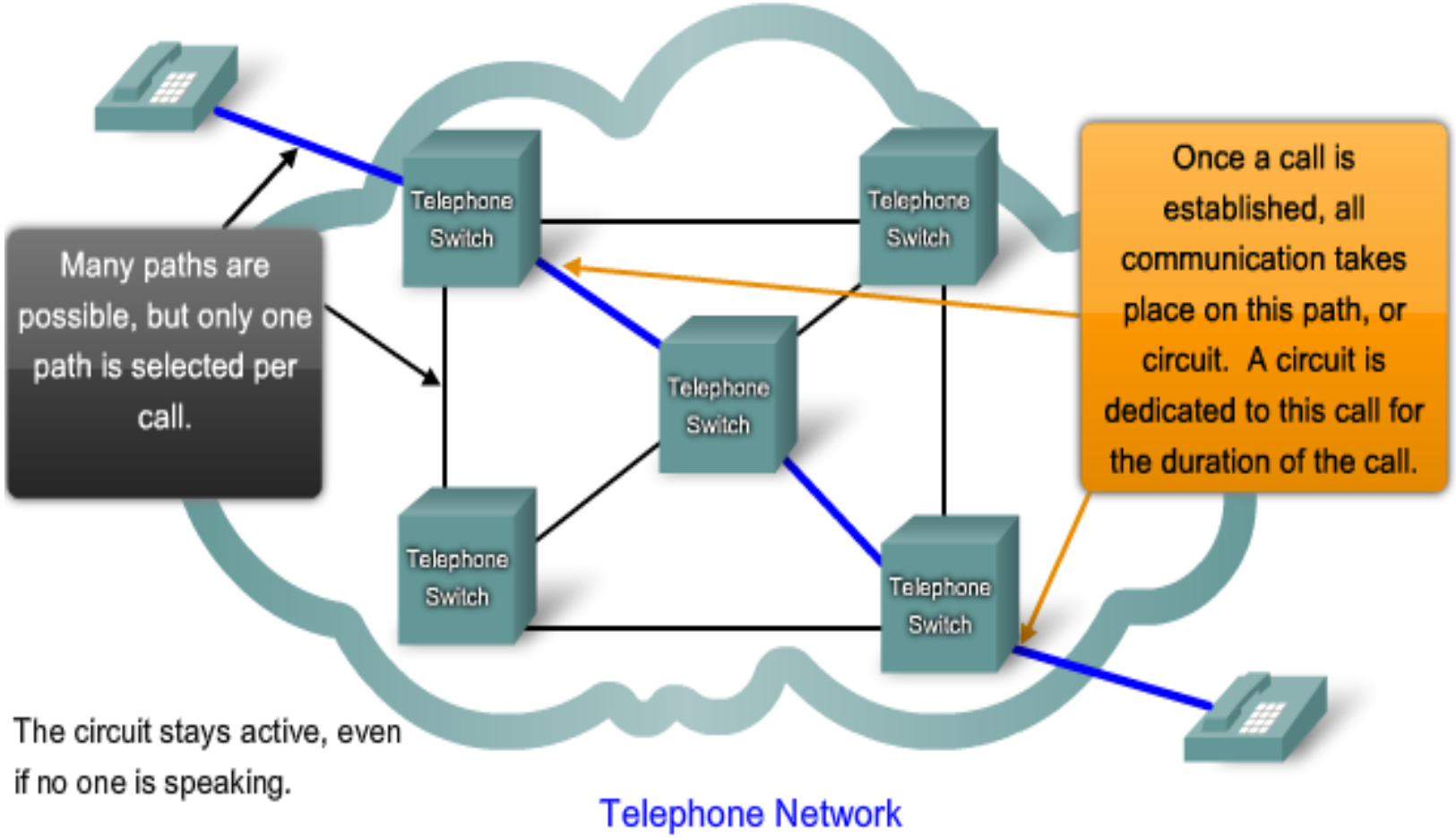
Circuit Switching in a Telephone Network

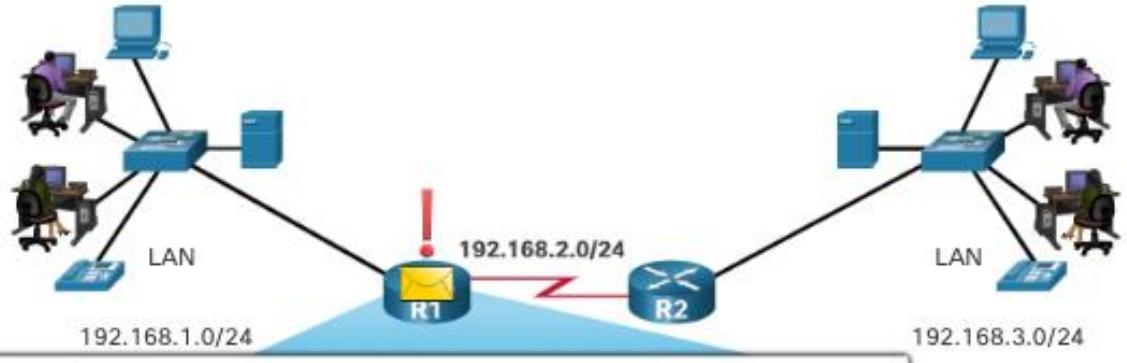


Functions of a Router

- When a router receives an IP packet on one interface, it determines which interface to use to forward the packet to the destination. This is known as routing.
- The interface that the router uses to forward the packet may be the final destination, or it may be a network connected to another router that is used to reach the destination network. Each network that a router connects to typically requires a separate interface, but this may not always be the case.
- The primary functions of a router are to determine the best path to forward packets based on the information in its routing table, and to forward packets toward their destination.

Functions of a Router

- The router uses its IP routing table to determine which path (route) to use to forward a packet. R1 and R2 will use their respective IP routing tables to first determine the best path, and then forward the packet.




```
R1# show ip route
Codes:
C - connected, S - static, I - IGRP, R - RIP, M - mobile, B -
BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default
U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, Serial0/0/0
S    192.168.3.0/24 [1/0] via 192.168.2.2
```

Routers use the routing table like a map to discover the best path for a given network.

Routing metrics

- Routing metrics and costs are used to determine the best path to the destination hop. Metrics are the network variables that can be used to determine the best path or route to the destination.
- Those routing metrics can be either static or dynamic routing.

The most common routing metrics are the following

Hop count: number of routing devices the packet needs to travel to reach the destination network.

Bandwidth: cumulative bandwidth of the link to the destination kilobytes per second.

Load: Load refers to the degree to which the network resource such as a router or network link is busy. A Load can be calculated in a variety of ways such as CPU utilization, packets processed per second.

Reliability: Reliability is a metric factor may be composed of a fixed value. It depends on the network links, and its value is measured dynamically. Some networks go down more often than others. Some network links repaired easily than other network links.

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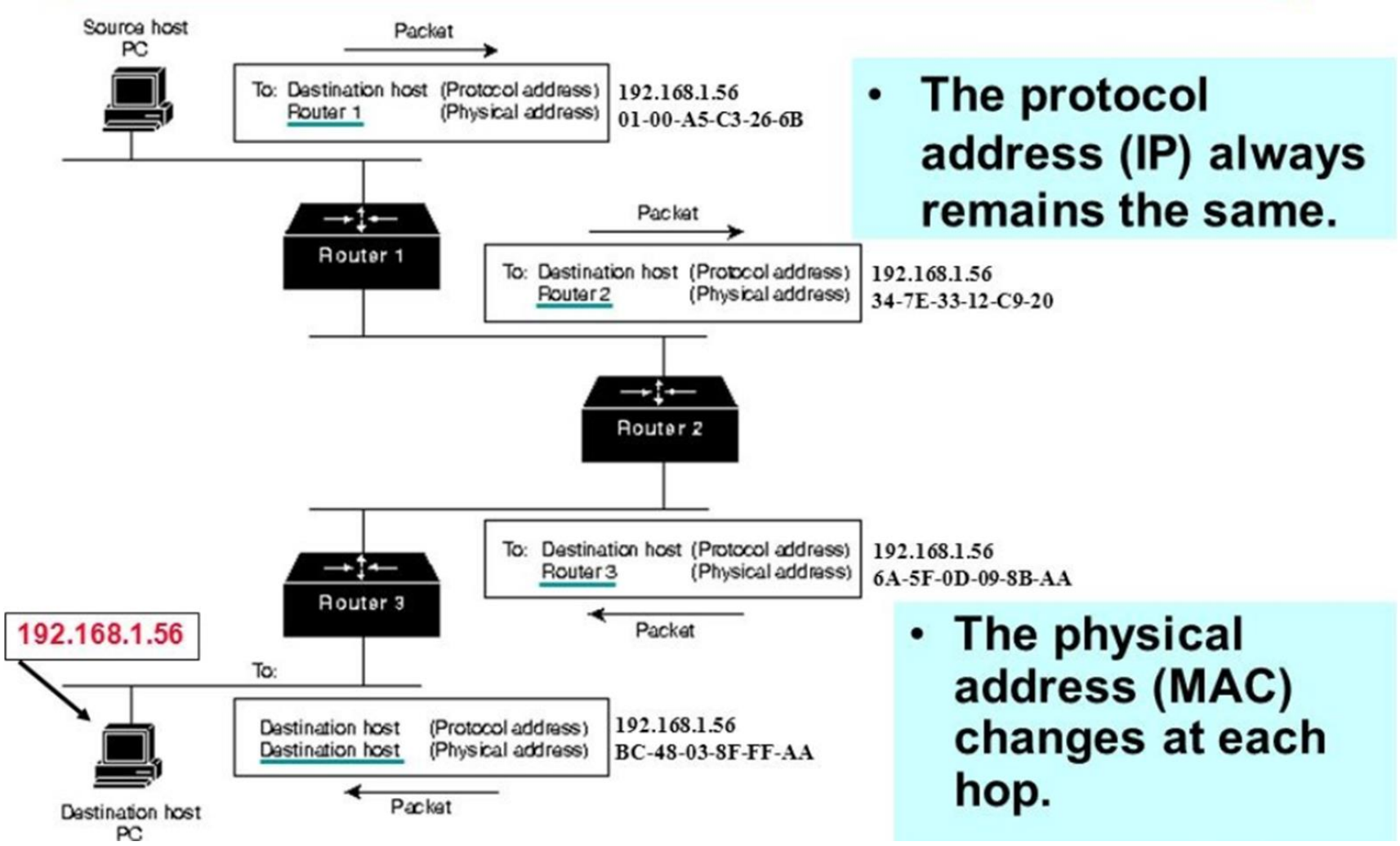
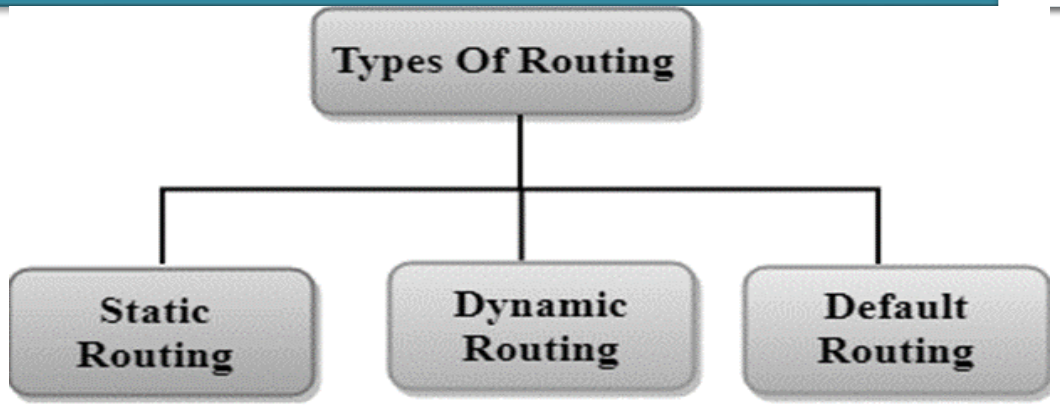
❖ Routing metrics and costs are used to determine the best path to the destination hop. Metrics are the network



- The protocol address (IP) always remains the same.
- The physical address (MAC) changes at each hop.



Static routing

- Static Routing is also known as Nonadaptive Routing.
- The administrator manually adds the routes in a routing table.
- A Router can send the packets for the destination along the route defined by the administrator.
- Routing decisions are not made based on the condition or topology of the networks
- Routes do not change unless changed by the administrator.

Advantages of static Routing

- Easy to implement in small network
- Very secure no advertisements are sent .
- Route to the destination is always the same unless the network administrator changes
- No routing algorithms or update mechanism required , therefore extra resource (CPU or RAM) are not required.

Disadvantage of Static route

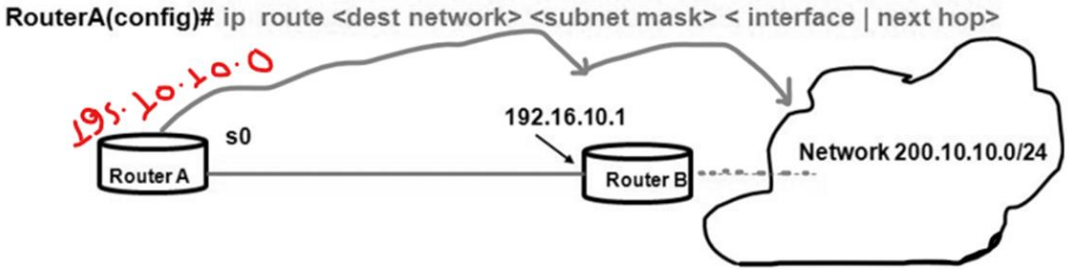
- For a large network, it becomes a very difficult task to add each route manually to the routing table.
- Configuration complexity increase as the network grows .
- It demands high skill from the network administrators.

Configuring static routes

- Use the following steps to configure static routes:
- Determine all desired destination networks, their subnet masks, and their gateways. A gateway can be either a local interface or a next hop address that leads to the desired destination.
- Enter global configuration mode.
- Type the ip route command with a destination address and subnet mask followed by their corresponding gateway from Step one. Including an administrative distance is optional.

Static routing

- Repeat Step three for as many destination networks as were defined in Step one.
- Exit global configuration mode.
- Save the active configuration to NVRAM by using the copy running-config startup-config command.
- Static routing example



- Packets arriving at router A with the network 200.10.10.0/24 as their destination will be sent out of router A through serial port S0. The static address command for network 195.10.10.0 to get to the network 200.10.10.0 is

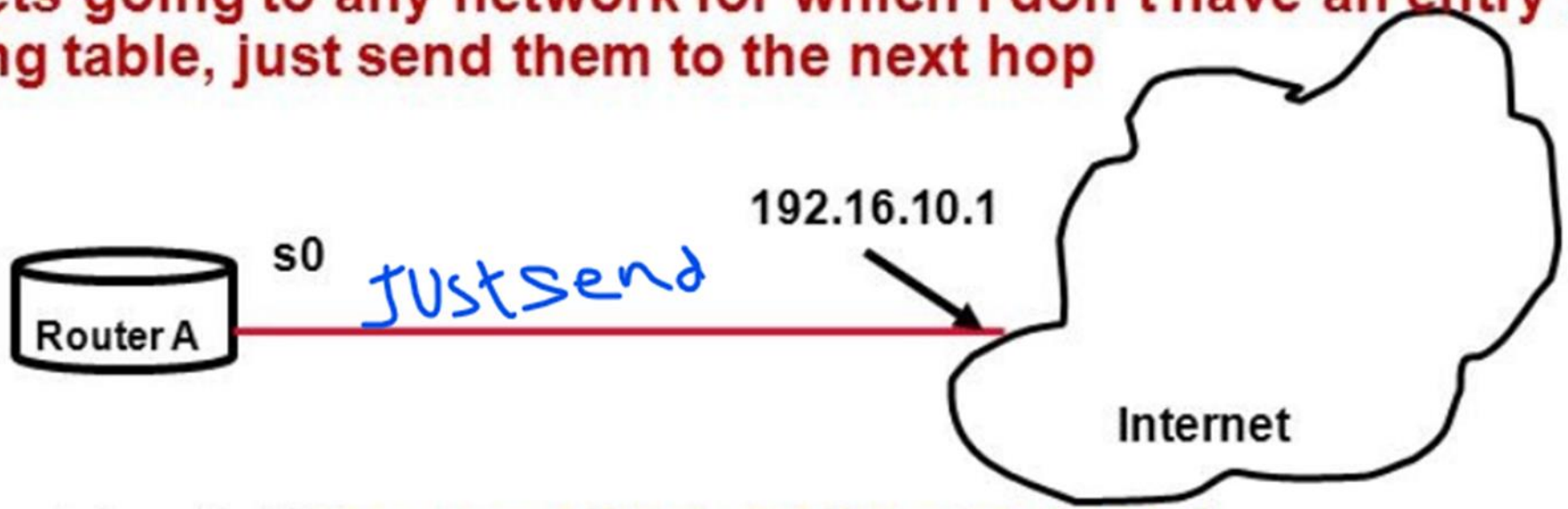
RouterA(Config) ip rout 200.10.10.0 255.255.255.0 S0 or

RouterA(Config) ip rout 200.10.10.0 255.255.255.0 192.16.10.1

Default Routing

❖ If the router doesn't know where to send the packets it uses default route .

Packets going to any network for which I don't have an entry in the routing table, just send them to the next hop



RouterA(config)# ip route 0.0.0.0 0.0.0.0 192.16.10.1

OR

RouterA(config)# ip route 0.0.0.0 0.0.0.0 s0

Dynamic Routing

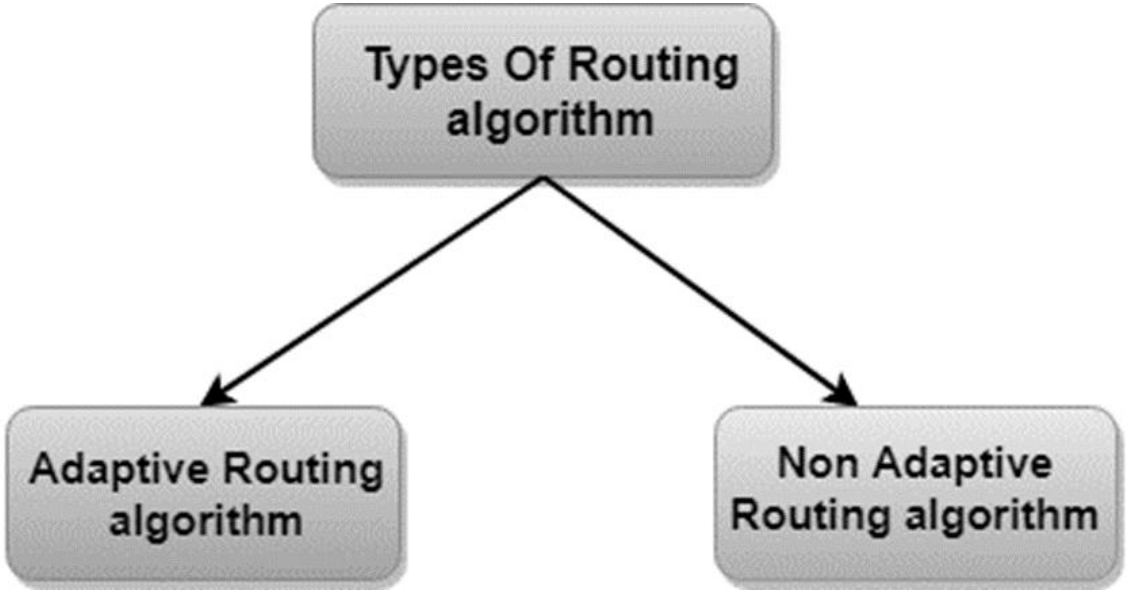
- It is also known as Adaptive Routing.
- It is a technique in which a router adds a new route in the routing table for each packet in response to the changes in the condition or topology of the network.
- Dynamic protocols are used to discover the new routes to reach the destination.
- In Dynamic Routing, RIP and OSPF are the protocols used to discover the new routes.
- If any route goes down, then the automatic adjustment will be made to reach the destination.
- The Dynamic protocol should have the following features:
- All the routers must have the same dynamic routing protocol in order to exchange the routes.
- If the router discovers any change in the condition or topology, then router broadcast this information to all other routers.

Advantages of Dynamic Routing:

- It is easier to configure.
- It is more effective in selecting the best route in response to the changes in the condition or topology.
- Disadvantage: it is more expensive in terms of cpu utilization and less secured as compared to others.

Routing Algorithm

- A routing algorithm is a procedure that lays down the route or path to transfer data packets from source to the destination.
- They help in directing Internet traffic efficiently. After a data packet leaves its source, it can choose among the many different paths to reach its destination.
- Routing algorithm mathematically computes the best path, i.e. "least – cost path" that the packet can be routed through.



Routing Algorithm

Adaptive routing algorithm /dynamic routing algorithm.

- This algorithm makes the routing decisions based on the topology and network traffic.
- The main parameters related to this algorithm are hop count, distance and estimated transit time

Non-Adaptive Routing algorithm

- Non Adaptive routing algorithm is also known as a static routing algorithm.
- When booting up the network, the routing information stores to the routers.
- Non Adaptive routing algorithms do not take the routing decision based on the network topology or network traffic.
- Examples of Non adaptive routing algorithm
  1. Flooding
  2. Random walks (RW)

Link state routing algorithms

Dijkstra's algorithm

- net topology, link costs known to all nodes
- accomplished via "link state broadcast"
- all nodes have same info
- computes least cost paths from one node ('source") to all other nodes
- gives forwarding table for that node

Notation:

- c(x,y): link cost from node x to y; = ∞ if not direct neighbors
- D(v): current value of cost of path from source to destination. v
- p(v): predecessor node along path from source to v
- N: set of nodes whose least cost path definitively known

Dijkstra's algorithm

```
Initialization:
  N = {S}
  for all nodes v
    if v adjacent to S
      then D(v) = c(S,v)
      else D(v) = infinity

Loop
  find w not in N such that D(w) is a minimum
  add w to N
  update D(v) for all v adjacent to w and not in N:
      D(v) = min( D(v), D(w) + c(w,v) )
  /* new cost to v is either old cost to v or known
   shortest path cost to w plus cost from w to v */
until all nodes in N
```

Notation:

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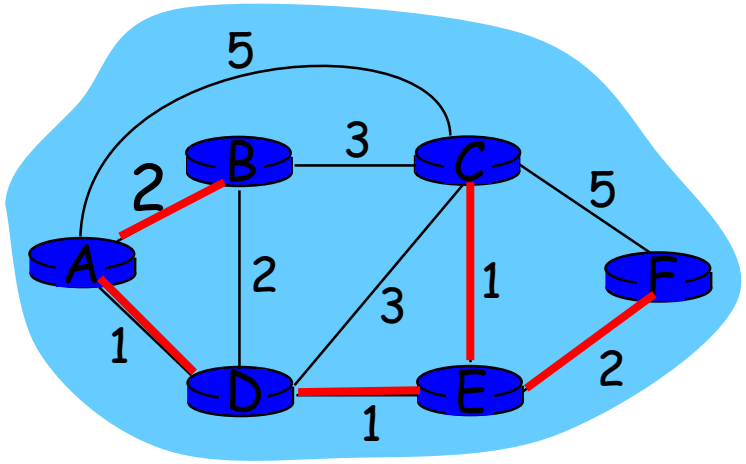
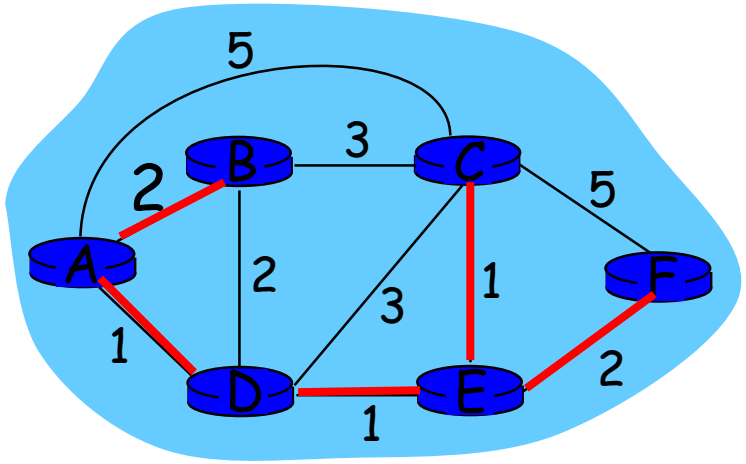
Dijkstra's algorithm: example

Step	start N	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	infinity	infinity
1	AD	2,A	4,D		2,D	infinity
2	ADE	2,A	3,E			4,E
3	ADEB		3,E			4,E
4	ADEBC					4,E
5	ADEBCF					



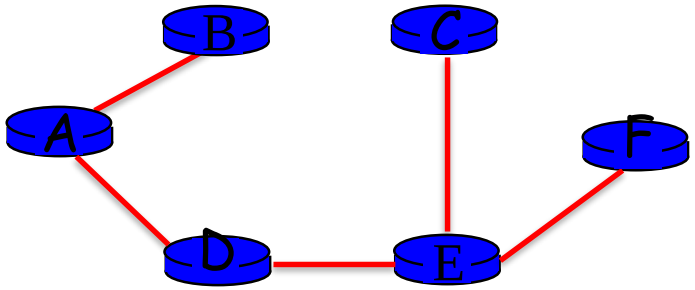
Dijkstra's algorithm: example

Step	start N	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	infinity	infinity
1	AD	2,A	4,D		2,D	infinity
2	ADE	2,A	3,E			4,E
3	ADEB		3,E			4,E
4	ADEBC					4,E
5	ADEBCF					



Dijkstra's algorithm: example…

Resulting shortest-path tree from A:



Resulting forwarding table in A:

destination	link
B	(A,B)
D	(A,D)
C	(A,D,E,C)
E	(A,D,E)
F	(A,D,E,F)

Distance Vector Algorithm

<u>Bellman-Ford Equation (dynamic programming)</u>

Define

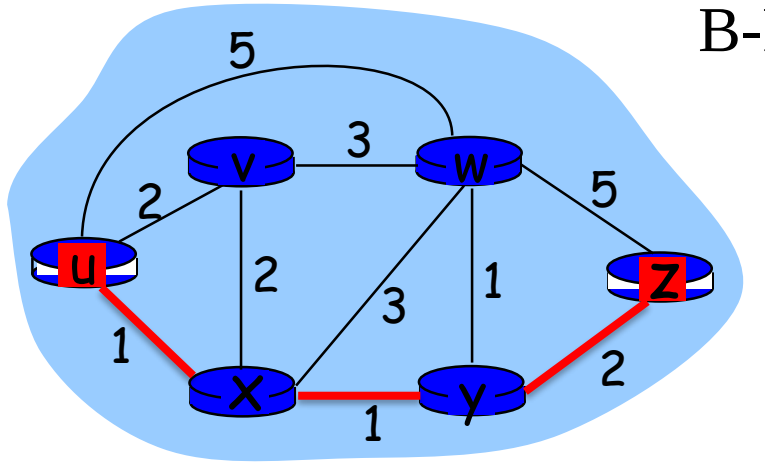
$d_x(y)$:= cost of least-cost path from x to y

Then

$$\boxed{d_x(y) = \min_v \{c(x,v) + d_v(y)\}}$$

where min is taken over all neighbors v of x

Bellman-Ford example



Clearly, $d_v(z) = 5$, $d_x(z) = 3$, $d_w(z) = 3$

B-F equation says:

$$d_u(z) = \min \{ c(u,v) + d_v(z),\ c(u,x) + d_x(z),\ c(u,w) + d_w(z) \}$$
$$= \min \{2 + 5,\ 1 + 3,\ 5 + 3\} = 4$$

Node that achieves minimum is next hop in shortest path ➜ forwarding table

dx(z)=min{(c(x,w)+dw(z),
c(x,v)+dv(z),
c(x,y)+dy(z)}
dv(z)=min{c(v,w)+dw(z),
c(v,x)+dx(z)
}

dx(z)=min{3+5,
2+8
3,}
dx(z)=3

Dw(z)=min{dw(z),
c(w,y)+dy(z)}
min{5,3)
dw(z)=3

Bellman-Ford example

Clearly, $d_v(z) = 5$, $d_x(z) = 3$, $d_w(z) = 3$

Here is the calculation

dx(z)=min{(c(x,w)+dw(z), c(x,v)+dv(z), c(x,y)+dy(z)} dx(z)=min{3+5, 2+8 3,} dx(z)=3
Dw(z)=min{dw(z), c(w,y)+dy(z)} min{5,3) dw(z)=3

Distance Vector Algorithm

Reading Assignment

Assignment take four nodes and use distance vector algorithm to calculate the shortest distance for each node.